Expected value and variance

October 16, 2026

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Definition of expected value

Definition 1 (Expected value of a discrete rv). Let X be a discrete rv with pmf f and $\operatorname{supp}(X) = \{w_1, w_2, \ldots, w_k\}$. Then, the expected value of X is:

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- We often refer to the expected value of X as the expectation of X or mean of X.

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Exercise 2. Is $\mathrm{E}(X)$ a rv or a constant?

A: A constant. We're just summing up numbers. Outcomes are numbers, probabilities are numbers. So $\mathrm{E}(X)$ is just a number.

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$$= 3.5. \tag{5}$$

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    a. i.e., it's not necessarily true that $\mathrm{E}(X) \in \mathrm{supp}(X)$.
3. The expected value of a rv is not actually the value we "expect" to get.

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We already know that $\mathrm{E}(X)$ is the long-run mean of a rv X. But there is another reason we call $\mathrm{E}(X)$ the "mean" of X.

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So $\mathrm{E}(X)$ is the mean of the population .

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Recall that expected value (for a discrete rv) is defined in terms of summations so many of the properties of expectation come from properties of summation.

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A: Yes, by Property 2 that $\mathrm{E}(aX + b) = a\mathrm{E}(X) + b$. What are a and b in this case? $a = \frac{1}{c}, b = 0$.

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Note that we did not need the pmf of R to evaluate $\mathrm{E}(R)$.

Useful theorem for calculating expected values

Theorem 1. Let X be a discrete rv with $\operatorname{supp}(X) = \{w_1, \ldots, w_k\}$ and pmf f_X. Let $Y := g(X)$, where $g : \mathbb{R} \to \mathbb{R}$. Then,

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Exercise 9. Let D be the number from rolling a fair 6-sided die. Evaluate $\mathrm{E}(D^2)$.

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Exercise 9. Let D be the number from rolling a fair 6-sided die. Evaluate $\mathrm{E}(D^2)$.

Hint: $g(x) = x^2$. Now, plug in g into the formula of Theorem 1.

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- Is $\mathrm{V}(X)$ a constant or a rv? A constant, since it's an expected value.

Equivalent expression for variance

Recall that we proved an equivalent expression for the sum of squared deviation from the mean of a sequence, x_i:

$$\sum_{i=1}^{n}(x_i - \overline{x})^2 = \sum_{i=1}^{n} x_i^2 - n(\overline{x})^2.$$

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Theorem 2. For any rv X,

$$\mathrm{V}(X) = \mathrm{E}\left[(X - \mu_X)^2\right] = \mathrm{E}(X^2) - \mu_X^2.$$

Proof that $\mathrm{E}\left[(X-\mu_X)^2\right] = \mathrm{E}(X^2) - \mu_X^2$

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- (4) -> (5) combines terms.

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Recall the example where D is the roll from a fair 6-sided die.

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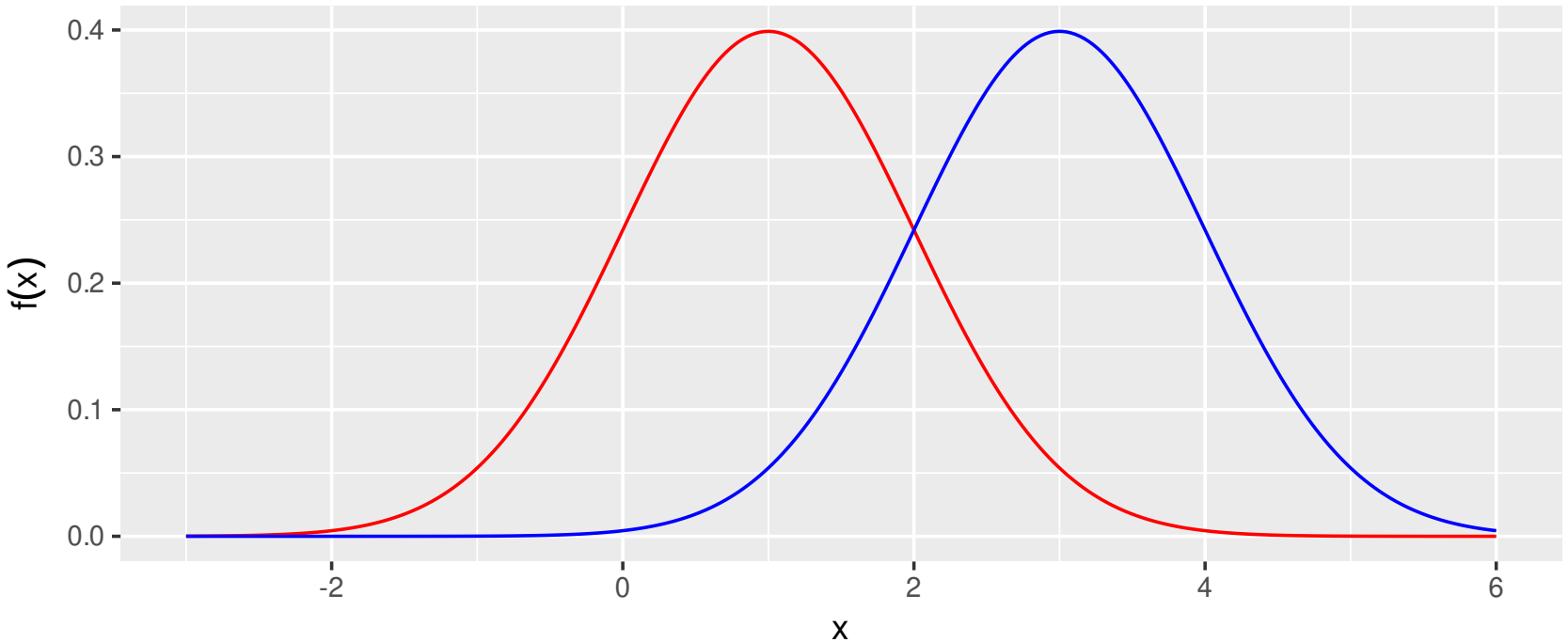
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Visualizing $\mathrm{V}(X + c)$



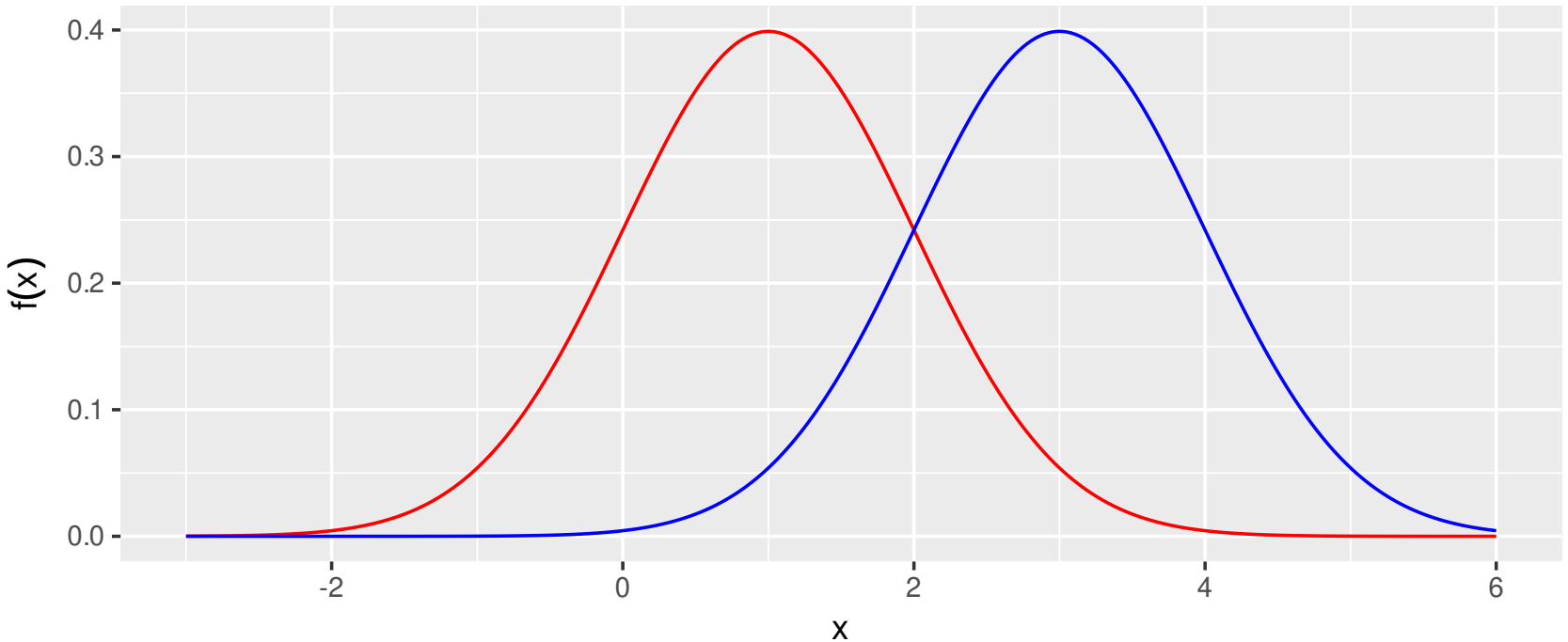
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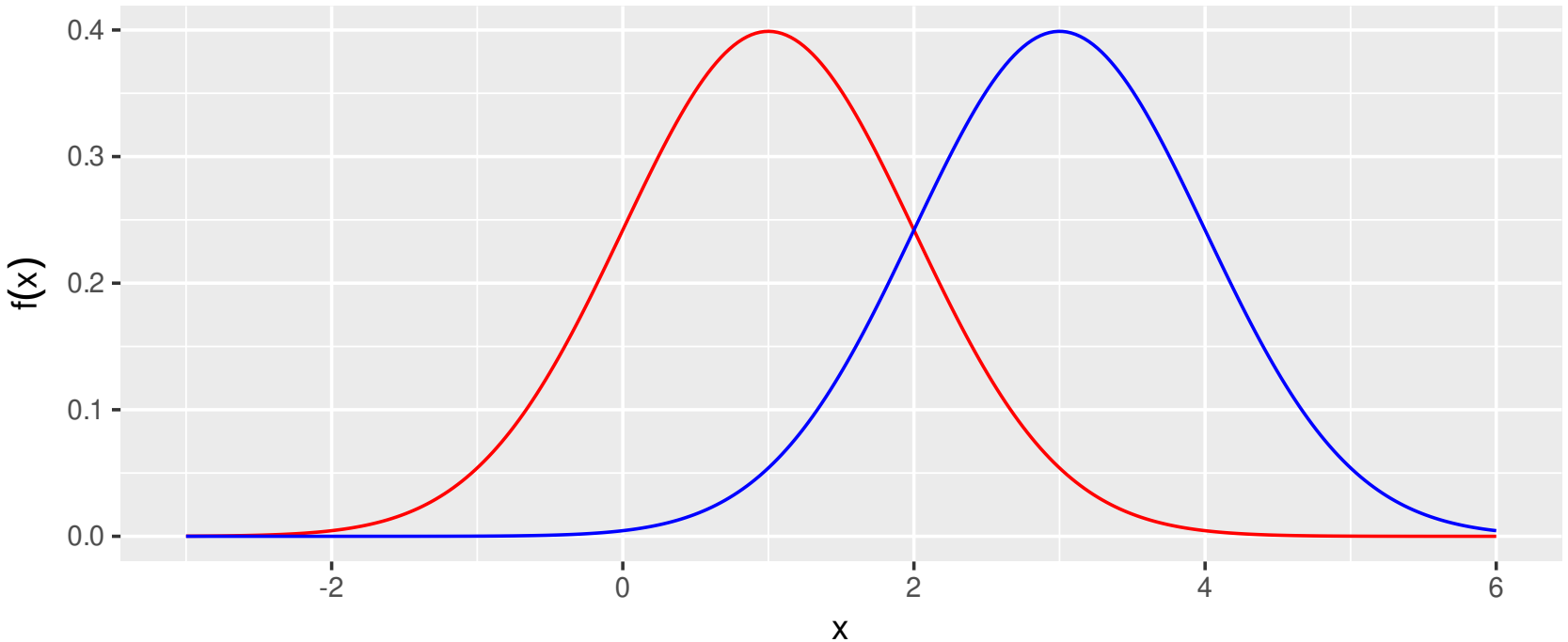
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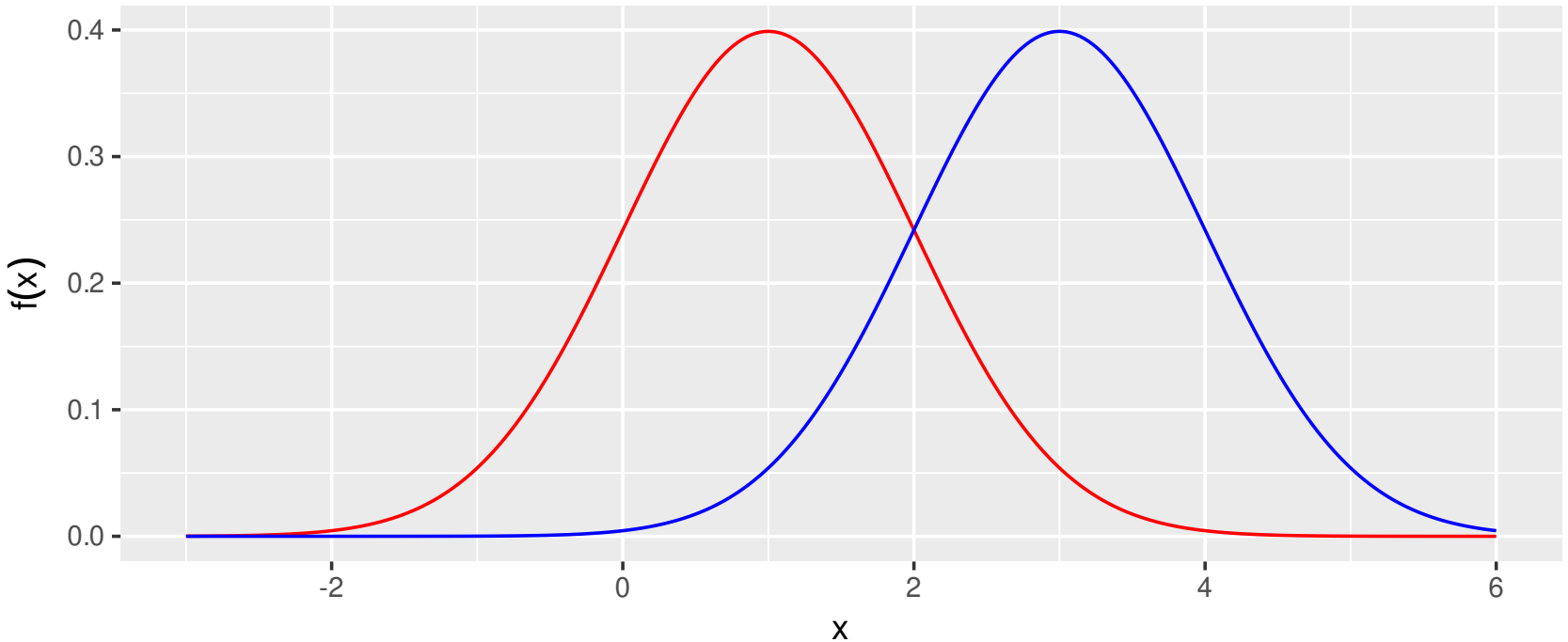
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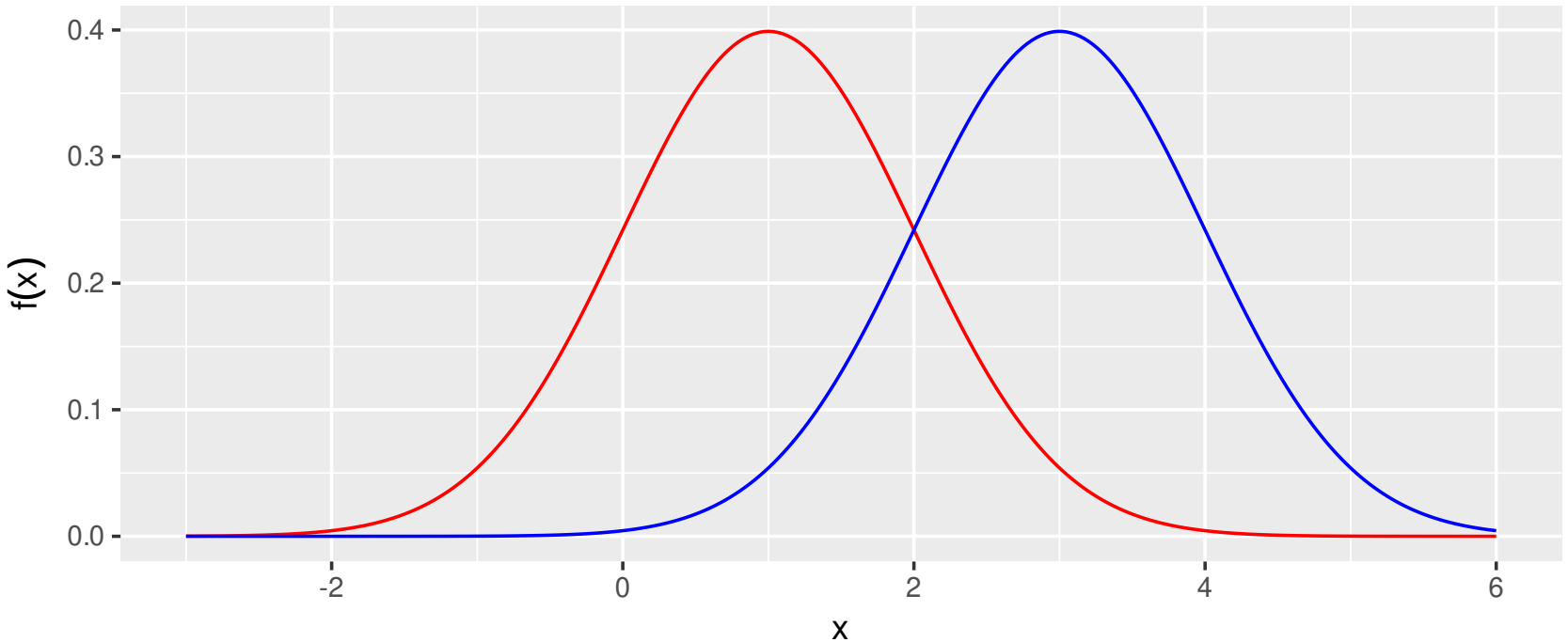


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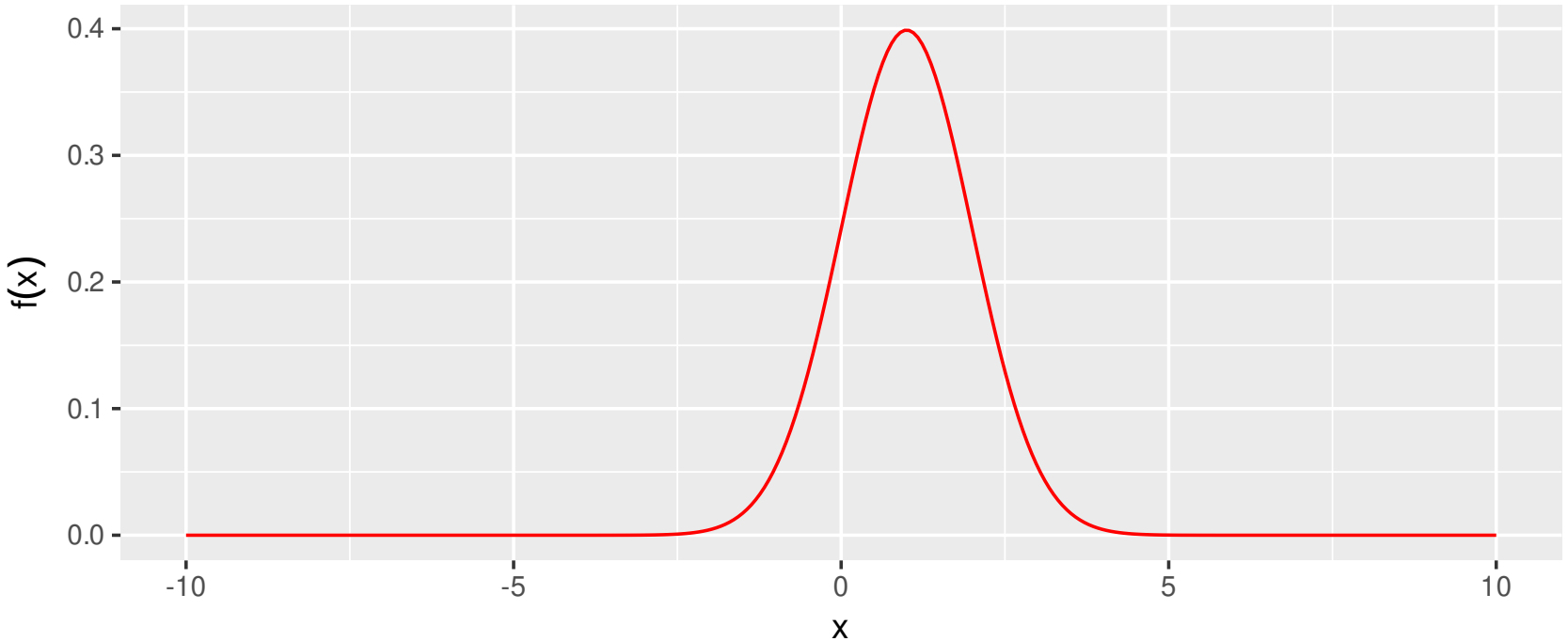


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Exercise 11. $\mathrm{E}(W) = \mathrm{E}(X + 2) = \mathrm{E}(X) + \mathrm{E}(2) = 1 + 2 = 3.$

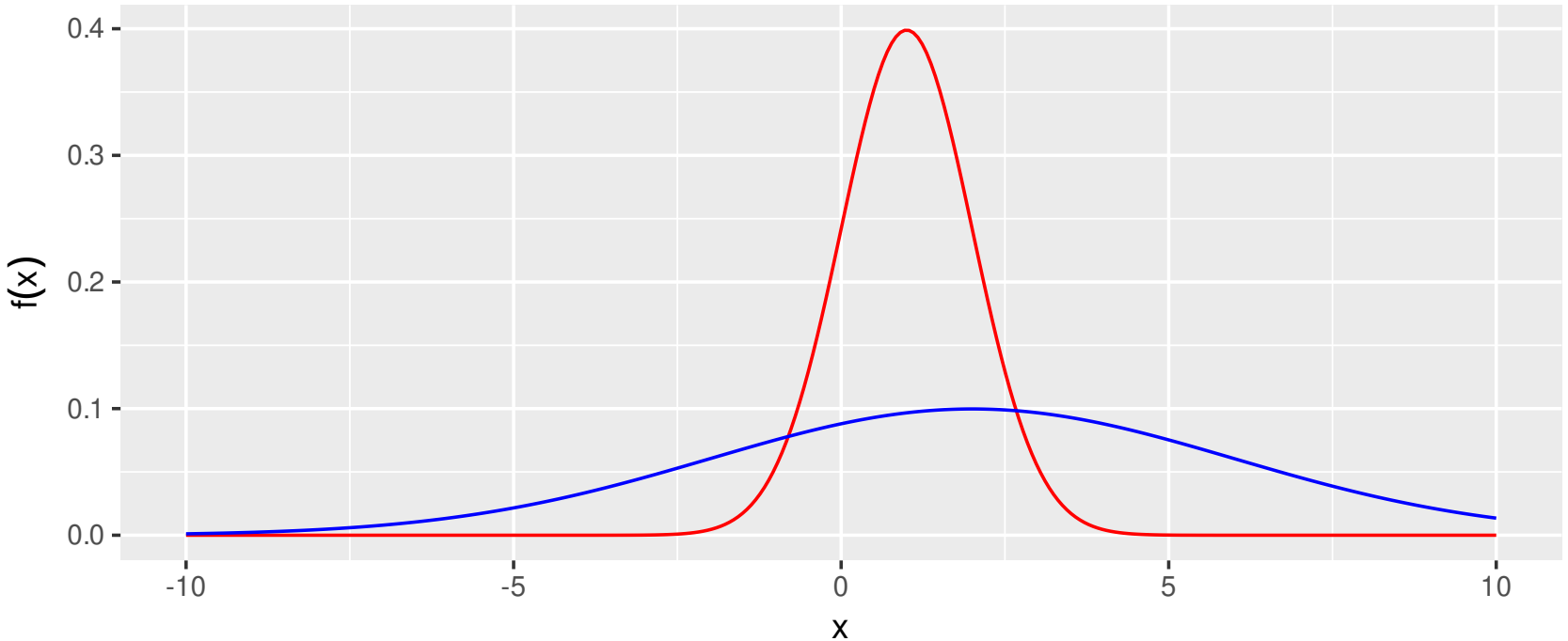
Exercise 12. $\mathrm{V}(W) = \mathrm{V}(X + 2) = \mathrm{V}(X) = 1.$

Visualizing $\mathrm{V}(cX) = c^2\mathrm{V}(X)$



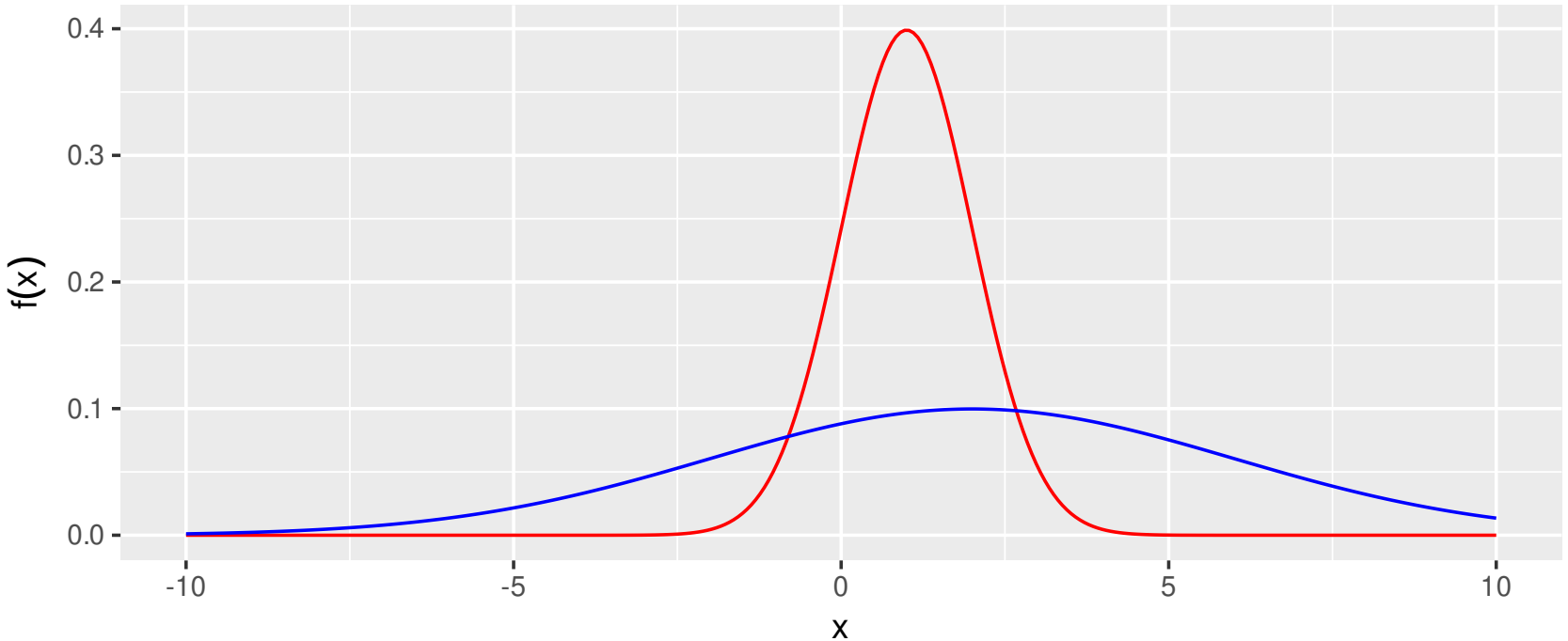
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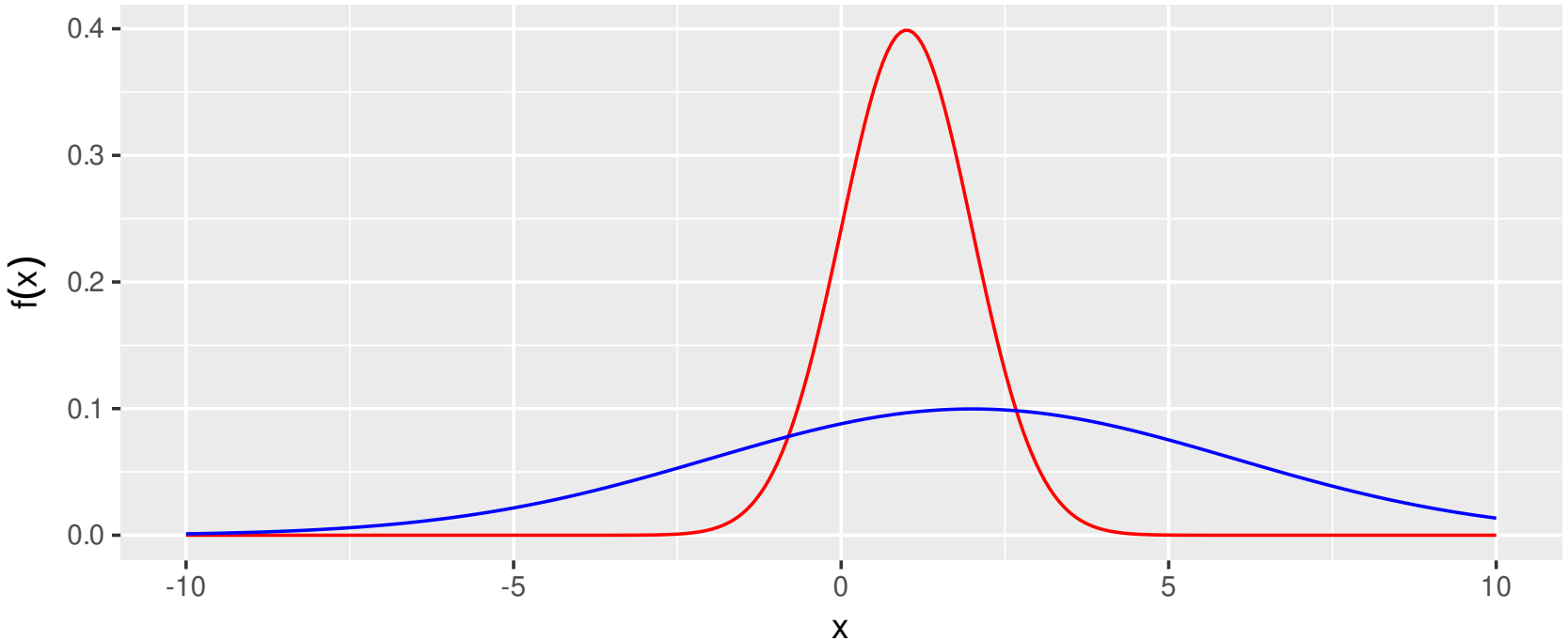
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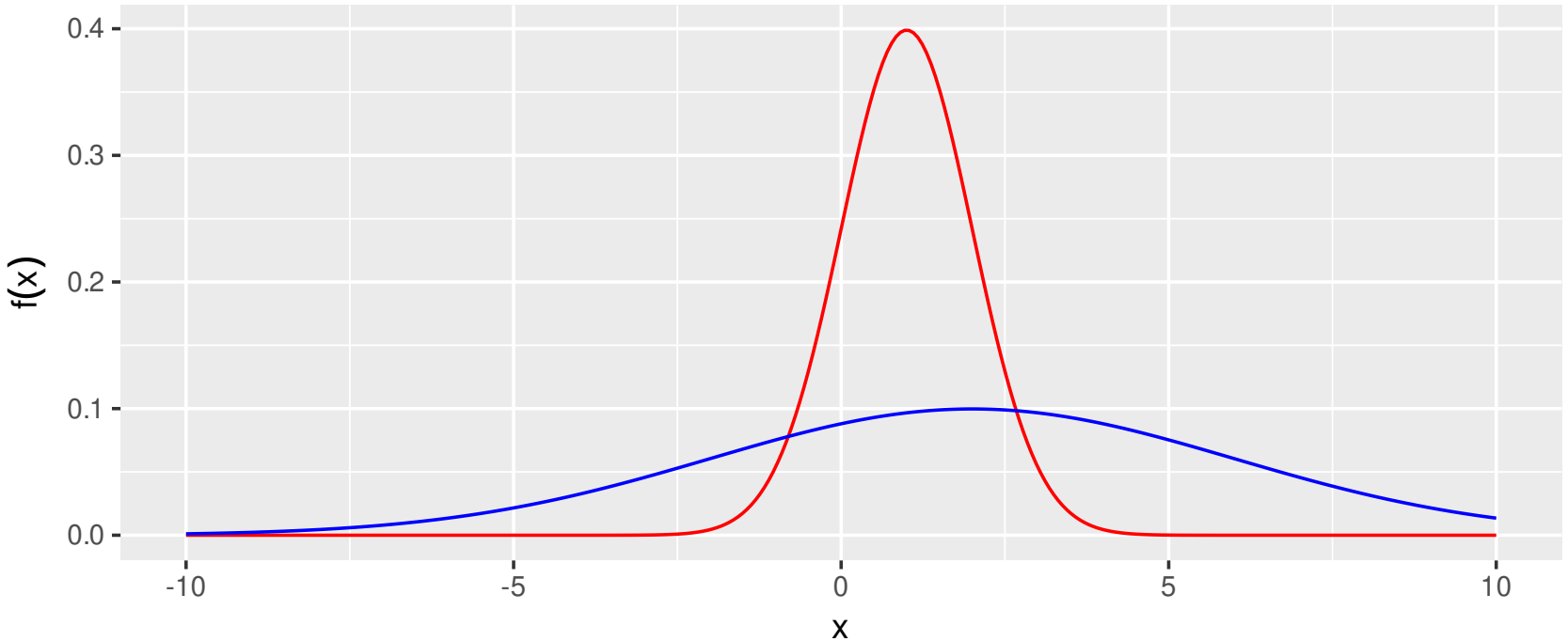
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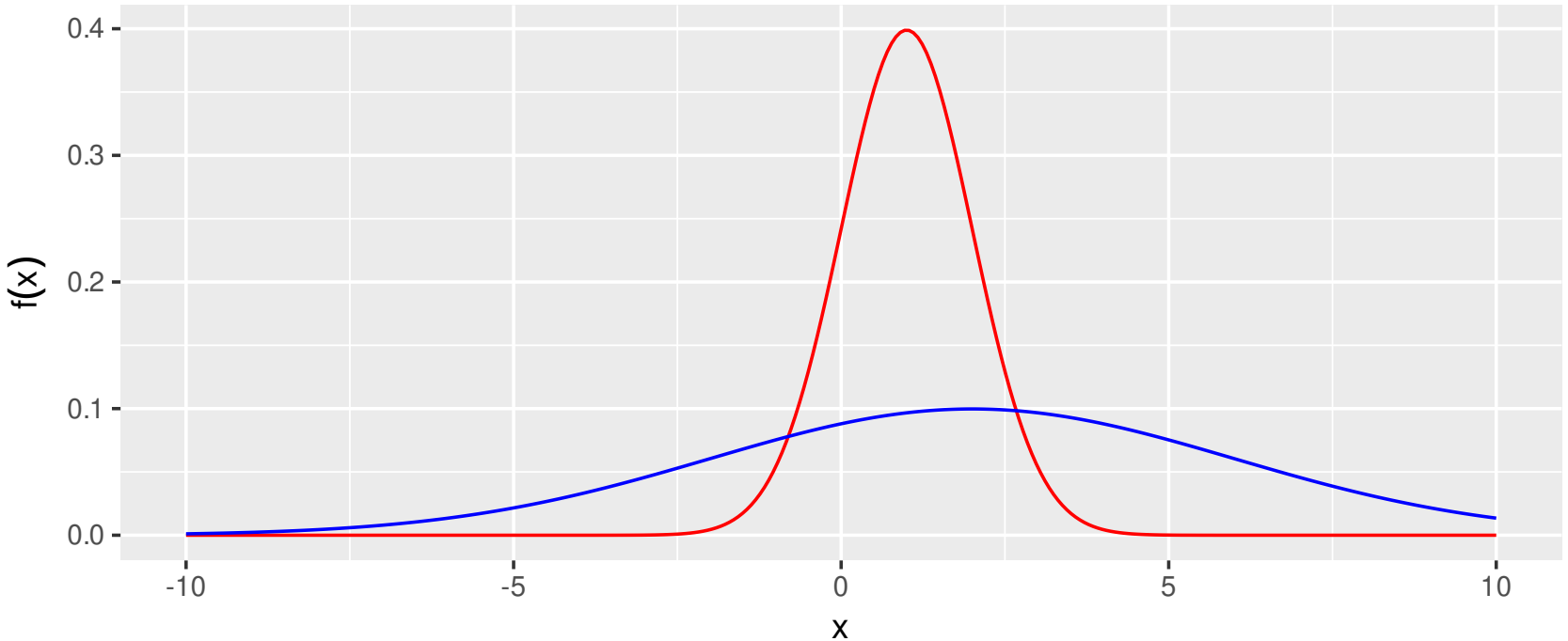


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Standard deviation

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Common notation is $\sigma_X := \mathrm{SD}(X)$.

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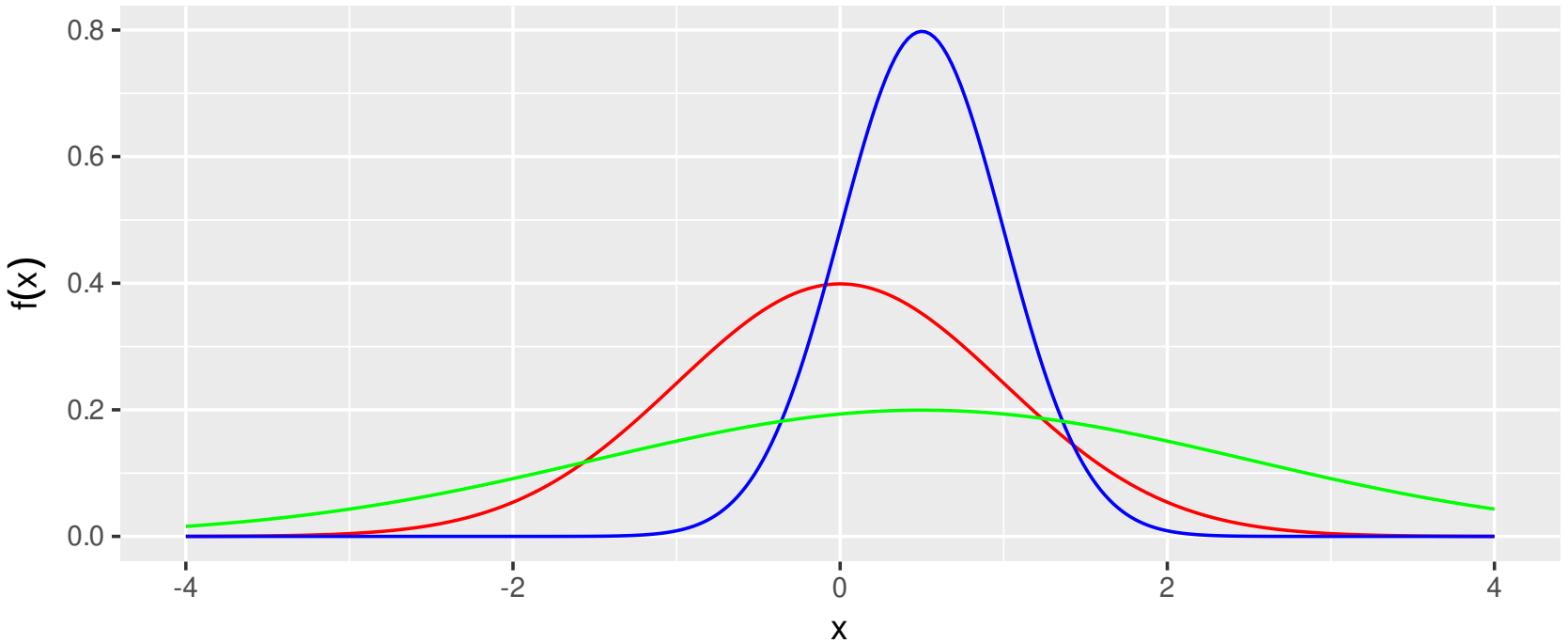
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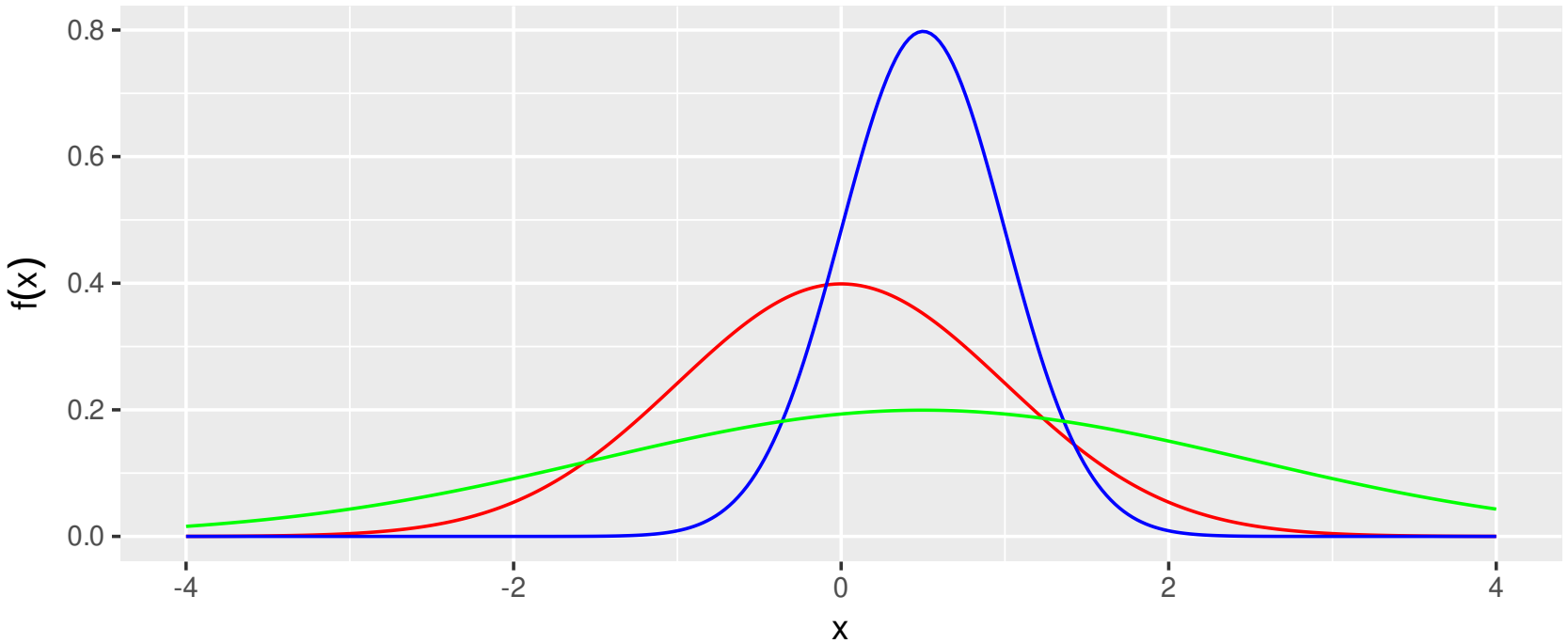
Notation 1. We use $\sim$ to mean "has the distribution". e.g., $X \sim N(0, 1)$ means "X has the standard normal distribution."

Name that normal



Exercise 16. Densities of the following three distributions are shown above: $N(0,1)$, $N(\frac{1}{2},\frac{1}{4})$, $N(\frac{1}{2},4)$. Which is which?

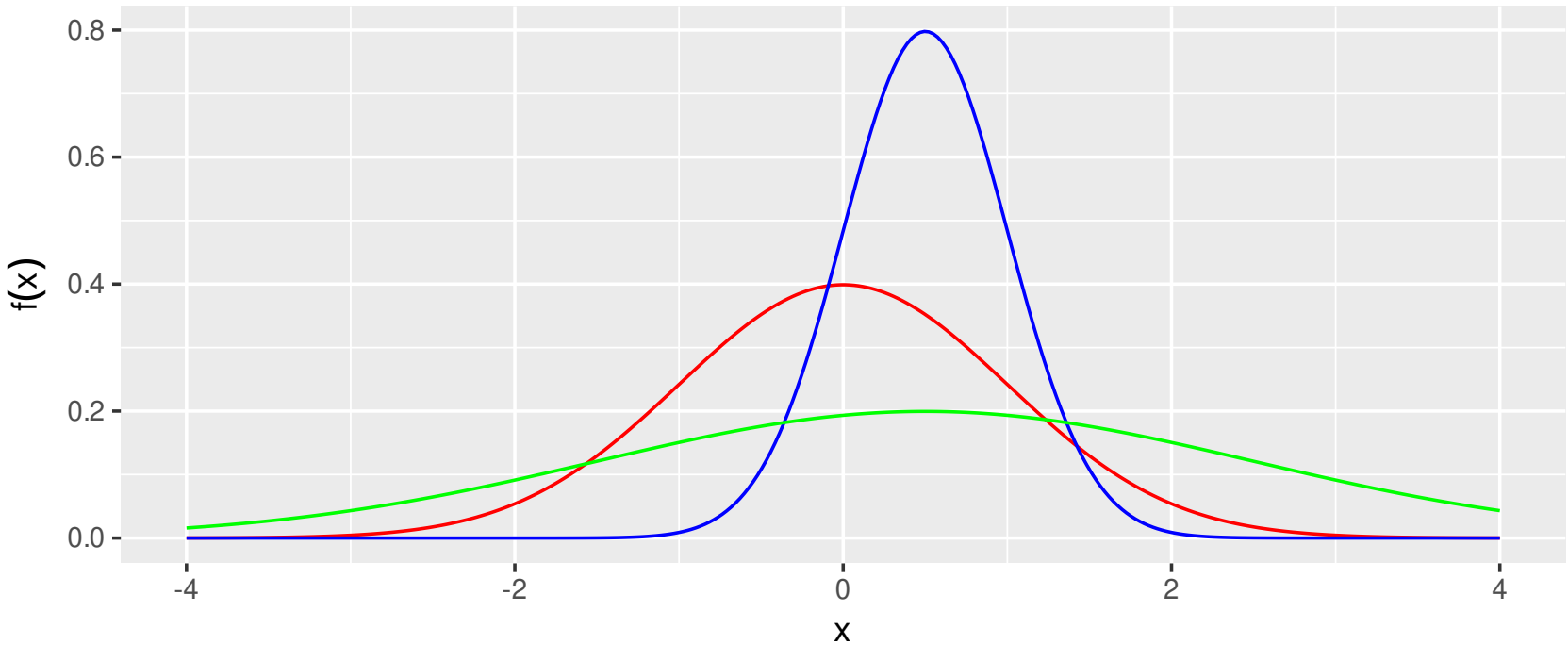
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